

Kate Y. L. Su

Curriculum Vitae

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Research Interests

- **Debris Disk Evolution** Investigating the dust emission around stars at different ages from main-sequence stars to white dwarfs to further understand the origin, evolution and fate of our Solar System via modeling the distributions and properties of the dust in circumstellar disks using spatially resolved imaging data and/or global spectral energy distributions.
- **Planetary Systems** Investigating the implication of debris disks on planet formation and evolution and understanding the observational characteristics of debris disks as tracers of planet formation.

Education

- 1996–2000 **Ph.D. in Astrophysics**, University of Calgary, Alberta, Canada.
1993–1995 **M.Sc. in Astronomy**, National Central University, Taiwan.
1989–1993 **B.Sc. in Physics**, National Central University, Taiwan.

Professional Experiences

- 2024–Present **Senior Scientist**, Space Science Institute, off-site AZ.
2017–Present **Astronomer**, Steward Observatory, University of Arizona, AZ.
2011–2017 **Associate Astronomer**, Steward Observatory, University of Arizona, AZ.
2013–2017 **Visiting Scholar**, ASIAA, Taiwan.
2005–2011 **Assistant Astronomer**, Steward Observatory, University of Arizona, AZ.
2001–2005 **Post-Doc.**, Steward Observatory, University of Arizona, AZ.
2000–2001 **Post-Doc.**, Dept. of Physics & Astronomy Valparaiso University, IN.

Professional Services

- 2018–Present **Member of the International Astronomical Union.**
2024 **SOC for the Dust Devils Workshop EVILS: Debris Disks in the Sonoran Desert.**
2018–2021 **Member of the American Astronomical Society.**
2018–2021 **Member of NRAO User Committee (UC) and ALMA North America Science Advisory Committee (ANASAC).**

- 2016–2019 **Member of the Science and Technology Definition Team (STDT) of the Origins Space Telescope.**
- 2015–2019 **Member of the SOFIA User Group.**
- Referee for **ApJ, ApJL, ApJS, AJ, MNRAS, A&A, Nature.**
- Panelist for **Telescope Allocation Committee:**, *Spitzer, HST, NOAO, NASA/Keck (panelist and panel chair), JWST.*
- Panelist for **Fellowships:**, *NASA NPP, Arizona 51 Peg b, Arizona Strittmatter, NASA Hubble (panel chair).*
- Panelist for **Grants:**, *NASA Origins, XRP, and ADAP programs, NSF AAG.*

Research Grants

Over \$3M grant money as PI:

Active:

- 2023–2026 **NASA-JWST**, *GO-03189: The Nature of Mineralogical Dichotomy in Extreme Debris Disks*, \$224,652.
- 2023–2026 **NASA-JWST**, *GO-03271: Characterizing the End Stage of Exoplanetary Systems*, \$288,991.
- 2023–2026 **NASA-JWST**, *GO-03786: Sizing up silicates at six small stars: characterising planetary debris orbiting white dwarfs*, \$183,923.
- 2023–2025 **NASA-JWST**, *GO-03690: The rest of the iceberg - dusty white dwarfs exposed by JWST*, \$90,705.
- 2022–2024 **NASA-JWST**, *GO-01647: Caught in the act: a debris disc in outburst*, \$188,218.
- 2022–2023 **NASA-XRP**, *Characterizing Rejuvenated Exoplanetary Systems – A Comprehensive View of Dusty White Dwarfs Using Archival Spitzer Data*, \$261,786.
- 2020–2023 **NASA-ADAP**, *Debris Disk Variability – Exploring the Diverse Outcomes of Large Collisions during the Eras of Oligarchic and Chaotic Growth II*, \$412,266.

Past:

- 2021–2022 **NASA-SOFIA**, *Constraining Planetesimal Collisions in the Terrestrial Planet Zone of HD 166191*, \$40,000.
- 2021–2022 **NASA-SOFIA**, *The Inner Warm Debris in the Vega Planetary System*, \$86,000.
- 2017–2019 **NASA-ADAP**, *Debris Disk Variability – Exploring the Diverse Outcomes of Large Collisions during the Eras of Oligarchic and Chaotic Growth I*, \$198,133.

- 2017–2019 **NASA-SOFIA**, *Mineralogical Evolution in Extreme Debris Disks II*, \$32,000.
- 2016–2018 **NASA-SOFIA**, *Mineralogical Evolution in Extreme Debris Disks I*, \$38,000.
- 2016–2018 **NASA-SOFIA**, *Structure of the Iconic Vega Debris Disk*, \$53,000.
- 2015–2018 **NASA-ADAP**, *The Outermost Planetesimal Zones in Extrasolar Debris Disks*, \$265,791.
- 2015–2017 **NASA-Spitzer**, *Debris Disk Variability: Observational Test Bed for Probing Terrestrial Planet Formation*, \$10,000.
- 2014–2017 **NASA-SOFIA**, *Nature of the Warm Excess in eps Eri: Asteroid Belt or Dragged-in Grains*, \$96,000.
- 2012–2013 **NASA-ADAP**, *Planetary System Evolution in the Terrestrial Zone*, \$212,119.
- 2011–2012 **NASA-Herschel**, *Herschel Observations of Solar-type Stars with Planets, Planetesimals and Dust*, \$22,000.
- 2011–2012 **NASA-Herschel**, *Dusty Disks around Hot White Dwarfs: Debris Disks or Remnant of post-AGB Binary Disks?*, \$47,619.
- 2010–2012 **NASA-Spitzer**, *A Complete Census of Warm Debris Disks*, \$106,750.
- 2010–2011 **NASA-HST**, *STIS Coronagraphic Imaging of the Kuiper Belt Surrounding the HR 8799 Planetary System*, \$10,000.
- 2009–2011 **NASA-ADAP**, *The Master Sample of Spitzer Debris Disk Measurements*, \$330,000.
- 2009–2011 **NASA-ADAP**, *A Photometry Catalog of Stars Within 130 pc*, \$40,000.
- 2008–2009 **NASA-Spitzer**, *Precise Ages for Debris Disk Stars via Gyrochronology*, \$20,000.
- 2008–2009 **NASA-Spitzer**, *The Nature of the 24 Micron Excess Emission from Hot White Dwarfs*, \$20,000.
- 2007–2007 **NASA-Keck**, *The Whole Picture: Characterizing Debris Disks with KIN, Spitzer and BLINC*, \$35,000.
- 2007–2008 **NASA-Spitzer**, *MIPS Survey of Dust Disks around Hot White Dwarfs*, \$15,000.
- 2006–2007 **NASA-Spitzer**, *The infrared Light Echo of V838 Monocerotis*, \$35,800.
- 2006–2007 **NASA-Spitzer**, *The Nature of the Dust around Lambda Bootis Stars: Debris Disks or Reflection Nebulae*, \$11,400.
- 2005–2006 **NASA-Spitzer**, *Deep 24 Micron Imaging of Nearby Debris Disks Around A Stars*, \$34,700.

- 2005–2006 **NASA-Spitzer**, *Infrared Studies of V838 Mon and V4332 Sgr – A New Class of Nova-like Variables*, \$16,581.
- 2005–2006 **NASA-Spitzer**, *Infrared Studies of the Helium Nova V445 Puppis*, \$10,785.
- 2005–2006 **NASA-Spitzer**, *Study of A New Class of Nova-like Variables*, \$20,700.
- 2002–2003 **NASA-HST**, *H₂ Imaging of Proto-Planetary Nebulae: Probing the Dynamics and Morphology*, \$10,456.

Invited Colloquia/Reviews in the Past 5 Years

- Jul 2021 **Tracing the Formation and Evolution of Planetary Debris Disks with Multiwavelength Approaches**, *SOFIA+ALMA joint Summer seminar series – exploring synergies between radio/millimeter and infrared observations*, Online webinar, <https://www.sofia.usra.edu/meetings-and-events/events/tracing-formation-and-evolution-planetary-debris-disks-multiwavelength>.
- Feb 2020 **Debris Disk Variability – A Legacy from Spitzer**, *Celebrating the Legacy of the Spitzer Space Telescope conference*, Pasadena, CA.
- May 2019 **Extreme Debris Disk Variability – a Tracer for the Impacts of Large Asteroids**, *New Horizons in Planetary Systems conference*, Victoria, BC, Canada.
- Oct 2018 **Probing Terrestrial Planet Formation by Witnessing Large Impacts in Extreme Debris Disks**, *Bombardment: Shaping Planetary Surfaces and Their Environments conference*, Flagstaff, AZ.
- Aug 2018 **Nano Dust as a Possible Cause of Hot Emission in Planetary Debris Disks**, *Astronomy in Focus XXX, presented at IAU XXX General Assembly*, Vienna, Austria.
- Jan 2018 **Circumstellar Disks in Young and Old Main-Sequence Stars**, *the 231st AAS Meeting, FIR-SIG Special Section*, DC.
- Nov 2017 **Detection and Characterization of Exo-Asteroid Belts**, *colloquium in Iowa State University*, IW.
- Jun 2017 **The Inner 25 au Debris Distribution in the eps Eri System**, *SOFIA Tele-talk*, NASA Ames, CA.
- Apr 2017 **Debris Disks: Tracers of Planet Formation**, *colloquium in UT-Austin*, TX.
- Sep 2016 **Debris Disks: Tracers of Planet Formation**, *colloquium*, SAO/CfA, Boston, MA.
- Sep 2016 **Detection and Characterization of Exo-Asteroid Belts**, *the Mike Jura Memorial Symposium*, Los Angeles, CA.

- Aug 2016 **Debris Disks and the FIR Surveyor: New insights into the formation and evolution of planetary systems**, *webinar for the Far-IR Science Interest Group*.
- Feb 2016 **Hot Dust Around Young and Evolved Stars**, *Astrophysics with the SPHEREx All-Sky Spectral Survey, SPHEREx Science Community Workshop*, Pasadena, CA.

Observing Programs

- JWST **GO-03189**: The Nature of Mineralogical Dichotomy in Extreme Debris Disks, xx hours.
- JWST **GO-03271**: Characterizing the End Stage of Exoplanetary Systems, xx hours.
- JWST **GO-03786**: Sizing up silicates at six small stars: characterising planetary debris orbiting white dwarfs, xx hours.
- JWST **GO-03690**: The rest of the iceberg - dusty white dwarfs exposed by JWST, Survey program
- JWST **GO-01647**: Caught in the act: a debris disc in outburst, 16.7 hours.
- Past Space Telescopes **2 HST, 11 Spitzer, 2 Herschel, and 6 SOFIA** programs (See past grants for details)
- ALMA PI program:
- 2013.1.00773.S**: Spatially Resolving the Planetesimal Disk around HD 95086 – A Young Analog of HR 8799
- 2019.1.01701.S**: Revealing the Debris Distribution in Two Rocky Planet Forming Disks
- 2021.1.00403.S**: The Origin of Warm Debris in Two-Temperature Debris Disks: Asteroidal or Cometary?
- Others NASA-Keck, Gemini-S, AZ-MMT, AZ-LBT, AZ-61", SMA, NASA-IRTF, ESO-VLT, GBT, LMT

Student Mentoring

- Yu-Chia Lin **Graduate, University of Arizona**, *Project title: "Optical Polarimetric Signature of Impact-Produced Dust Clump – A Case Study of RZ Psc"*, 1 paper published in 2023, 1-yr student in Department of Physics, 2023.
- Lewis Watt **Graduate, University of Bristol, UK**, *Project title: "Planetary embryo collisions and the wiggly nature of extreme debris discs"*, 1 paper published, graduated in 2023.

- Nicholas Ballering **Graduate, University of Arizona**, *Project title: “Measuring the structure and composition of circumstellar debris disks”*, 3 papers published, graduated in 2016, Current status: post-doc at University of Virginia.
- Sarah Morrison **Graduate, University of Arizona**, *Project title: “Multiple Planet-Debris Disk Interactions: Probing Planetary System Evolution”*, 1 paper published, graduated in 2017, Current status: Assistant Professor, Missouri State University.
- Vanessa Bailey **Graduate, University of Arizona**, *Project title: “Adaptive optics for directly imaging planetary systems”*, 2 papers published, graduated in 2015, Current status: WFIRST staff scientist at JPL.
- Tiffany Meshkat **Graduate, Leiden Observatory**, *Project title: “Searching for Planets in Holey Debris Disks”*, 2 papers published, graduated in 2015, Current status: WFIRST staff scientist at IPAC/Caltech.
- Huan Meng **Graduate, University of Arizona**, *Project title: “Planetary Collisions Outside the Solar System: Time Domain Characterization of Extreme Debris Disks”*, 3 papers published, graduated in 2015.
- Edward Montiel **Undergraduate, University of Arizona**, *Project title: “The Relation between Debris Disks and Stars with Very Low-Mass Companions Revealed through Ground-Based AO Imaging Surveys”*, PhD from LSU in 2016, Current status: staff scientist at SOFIA Science Center.
- Jana Bilikova **Graduate, UIUC**, *Project title: “Dusty Disks Around Hot White Dwarfs”*, 2 papers published, graduated in 2012.
- András Gáspár **Graduate, University of Arizona**, *Project title: “Observations and models of infrared debris disk signatures and their evolution”*, 2 papers published, graduated in 2011, Current status: assistant astronomer at University of Arizona.
- Nathan Stock **Graduate, University of Arizona**, *Project title: “The Structure of the β Leonis Debris Disk”*, 1 paper published.
- Jennifer Sierchio **Undergraduate Summer Student, University of Arizona**, *Project title: “Debris Disks around Solar-like Stars”*, 2 papers published, PhD from MIT in 2016.
- Juan Rafael Martínez Galarza **Undergraduate**, *Project title: “Modelling of Infrared Emission from λ Bootis Stars”*, 1 paper published, PhD from Leiden University in 2012, Current status: staff scientist at CfA.

Publications

- NASA ADS link: <https://ui.adsabs.harvard.edu/search/q=orcid%3A0000-0002-3532-5580>

- 22 first-authored peer-reviewed papers
- a total of 179 peer-reviewed papers
- total number of citations : over 12k
- H-Index for results: 57
- 7 Astro2020 white papers
- 13 press releases

Details can be found in the publication list.